NORMALIZATION OF SCHEMES COMMUTES WITH SMOOTH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5373-5386 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5388-5407, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Normalization of a scheme in a quasi-coherent sheaf of algebras, integral closure, and smooth base change; the reduction is retained.

Source reference. Original lemma coherent-lemma-flat-base-change-cohomology: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5407–5407. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma constructions-lemma-spec-properties: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5407–5407. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-integral-closure-smooth-pullback: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5359–5365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5407–5407. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-characterize-normalization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5407–5407. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1 (Normalization commutes with smooth base change)**.** *Let*

$$\begin{array}{ccc} Y_2 & \longrightarrow & Y_1 \\ {\scriptstyle f_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{\varphi} & X_1 \end{array}$$

be a fibre square in the category of schemes. Assume f_1 is quasi-compact and quasi-separated, and φ is smooth. Let $Y_i \to X_i' \to X_i$ be the normalization of X_i in Y_i. Then $X_2' \cong X_2 \times_{X_1} X_1'$.

Proof. The base change of the factorization $Y_1 \to X_1' \to X_1$ to X_2 is a factorization $Y_2 \to X_2 \times_{X_1} X_1' \to X_2$ and $X_2 \times_{X_1} X_1' \to X_2$ is integral (Morphisms, Lemma 01WL). Hence we get a morphism $h : X_2' \to X_2 \times_{X_1} X_1'$ by the universal property of Morphisms, Lemma morphisms-lemma-characterize-normalization. Observe that X_2' is the relative spectrum of the integral closure of $\mathcal{O}_{X_2}$ in $f_{2,*}\mathcal{O}_{Y_2}$. If $\mathcal{A}' \subset f_{1,*}\mathcal{O}_{Y_1}$ denotes the integral closure of $\mathcal{O}_{X_1}$, then $X_2 \times_{X_1} X_1'$ is the relative spectrum of $\varphi^*\mathcal{A}'$, see Constructions, Lemma constructions-lemma-spec-properties. By Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology we know that $f_{2,*}\mathcal{O}_{Y_2} = \varphi^* f_{1,*}\mathcal{O}_{Y_1}$. Hence the result follows from Lemma lemma-integral-closure-smooth-pullback. □